

Flight-Class Processor-in-the-Loop Validation of a Reliability-Asymmetric Spacecraft GNC Stack

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Abstract—A capable-but-certifiable spacecraft autonomy stack is only credible if it runs within real flight-compute budgets and behaves under the actuator and sensor non-idealities a clean simulator omits, not just in a desktop simulation. We report a flight-class validation of a reliability-asymmetric GNC stack (a learned fault-adaptive controller and a language-model commander, bounded by a Kind 2-verified runtime shield) on three measured fronts. The first is flight-class compute, taken to *processor-in-the-loop* on real silicon. On a Raspberry Pi 5 (quad Cortex-A76, the A53/A72-class stand-in the proposal called for), the controller+shield hot loop executes a cycle in $24.5\ \mu\text{s}$, $\approx 4000\times$ under the 100 ms 10 Hz budget, in 1.0 MB, and the int4 commander emits a grammar-valid plan at 28 tok/s in 437 MB (within the 512 MB cap), at $\approx 6.5\ \text{W}$ core-attributable power. This closes the emulation caveat on the silicon class that flies. The second is the runtime shield *run live in the closed loop*, which adds 0.02% overhead at 10 Hz while keeping a recovering controller 100% autonomous and escalating a diverging one every time. The third is a *high-fidelity fault campaign* that injects the non-idealities a physical platform would expose, namely real reaction-wheel momentum, saturation and friction, star-tracker and gyro noise, bias and quantisation, and finite actuator resolution and deadband. Across held-out faults the deployed recovery is statistically unchanged from the ideal case on the faults the estimator is designed for (100% autonomous, Wilson interval 96.9 to 100, against 0% for a fault-unaware baseline), and on faults beyond that scope, additive bias and a dead axis, the controller does not recover and the verified shield escalates every case, so safety holds exactly where autonomy is not earned. A physical air-bearing campaign is the natural next rung and is future work, and we map each fault onto its physical reaction-wheel manipulation to define it.

Index Terms—processor-in-the-loop, flight software, runtime assurance, embedded systems, spacecraft autonomy, fault injection

I. INTRODUCTION

A spacecraft autonomy stack earns flight only if its safety can be assured *and* it fits the vehicle’s compute, power, and timing budgets, and only if it behaves under real faults rather than idealized injections. A result that holds in a desktop simulator establishes none of these three things. The desktop machine is the wrong processor, so it speaks to neither the worst-case latency on a flight CPU nor the resident set that must coexist with the rest of the flight software in a few hundred megabytes. The injected fault is a clean multiplicative term on a perfectly modeled actuator, so it omits the encoder quantization, the bearing friction, the control-loop jitter, and the residual gravity torque that a real reaction wheel on a real platform brings. And the safety monitor, when it is timed

at all, is timed in isolation rather than inside the loop it is meant to protect, so its true cost while contending with the controller it watches is unknown. A claim that a learned-plus-verified GNC stack is *flyable* must therefore be retired from the simulator and re-established against flight-relevant bars. That re-establishment is the subject of this paper.

The stack under test is reliability-asymmetric. It pairs a learned fault-adaptive controller (a recurrent module in the rapid-adaptation style [2] infers an actuator fault online and feeds an analytic control law) with a compressed language-model commander, and it bounds both with a runtime shield of safety invariants whose predictor soundness is machine-checked by the Kind 2 model checker [3], in the run-time-assurance tradition for spacecraft control [4]. The organizing principle is that the tiers of an autonomy stack do not have equal verifiability, and the design assigns authority so that the element whose soundness can be *proved* bounds the elements whose soundness cannot. That principle is argued and evaluated in simulation in the companion work; the present paper asks the next question, which is whether the principle survives contact with real silicon and real hardware. Three properties have to hold for it to do so. The stack must run on the silicon class that flies, within the compute, memory, power, and timing budgets a flight-readiness review would impose. Its verified shield must stay cheap and correct *live* in the closed loop, not merely in a stand-alone microbenchmark. And the fault recovery the controller is built around must hold under the actuator and sensor non-idealities a clean simulator omits, where the idealizations of a perfectly modeled actuator and perfect sensing no longer apply.

We establish all three on measured evidence. The flight-class compute is measured on a real Raspberry Pi 5, not an emulator, which is the distinction that matters because a cycle-accurate QEMU model of a flight CPU still does not reproduce that CPU’s cache behavior, branch prediction, or memory bandwidth, and an autonomy stack benchmarked only under emulation carries a caveat a reviewer is right to press. The live-shield overhead is measured inside the integrated closed loop on the Basilisk attitude simulator [1], with the real machine-checked monitor evaluated every control step on the actual pointing state, so the number reported is the cost the shield actually imposes on the controller rather than an estimate of it. The third front answers the standing objection to any simulation result, that the actuator is perfectly modeled and the sensing is perfect, by re-running the same deployed stack through a high-fidelity model of the non-idealities a physical platform would expose, namely real reaction-wheel momentum, saturation and friction, star-tracker and gyro noise,

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bias and quantisation, and finite actuator resolution and a motor deadband, across held-out faults at a scale a single bench rig could not reach. A physical air-bearing campaign remains the established route to the effects that even a high-fidelity model leaves out, so we treat it as the next rung and as future work, and we map each fault onto its physical reaction-wheel manipulation so that the campaign is fully specified.

A. Contributions

- 1) We measure flight-class compute on real hardware (Sec. III). The hot loop and the commander are measured on a real Pi 5, against explicit 10 Hz timing, single-megabyte memory, 512 MB commander, and few-watt power budgets, closing the emulation caveat with measurements on the silicon class that flies.
- 2) We run the verified shield live and cheaply in the closed loop (Sec. IV). Run inside the closed loop on honest pointing state, the machine-checked monitor adds 0.02% overhead at 10 Hz while discriminating a recovering controller from a diverging one with a 100% versus 0% autonomy split.
- 3) We run a high-fidelity closed-loop fault campaign (Sec. V). We inject the actuator and sensor non-idealities a clean simulator omits, namely real reaction-wheel momentum, saturation and friction, star-tracker and gyro noise, bias and quantisation, and finite actuator resolution and a motor deadband, into the deployed closed loop, and show across held-out faults that recovery on the faults the estimator is designed for is statistically unchanged from the ideal case (100% autonomous, Wilson interval 96.9 to 100, against 0% for a fault-unaware baseline), while on faults beyond that scope the verified shield escalates every case.

II. RELATED WORK

This paper sits at the meeting point of four lines of work that bear directly on whether an onboard autonomy stack is flight-credible, namely hardware-in-the-loop validation for spacecraft GNC, edge and embedded inference for autonomy, runtime assurance in flight software, and physical attitude testbeds. We position the present validation against each, and the recurring point is that the individual practices are established while their *joint* application to a learned, adaptation-aware spacecraft stack on flight-class hardware is what is being demonstrated.

Hardware-in-the-loop validation for spacecraft GNC. Flight-software practice does not accept a GNC algorithm on the strength of desktop simulation alone. The standard path to flight progressively replaces modeled components with real ones, from software-in-the-loop, through processor-in-the-loop on the target CPU, to hardware-in-the-loop with real sensors and actuators in the control path, precisely so that timing, numerical, and actuator effects that a desktop model omits are exposed before flight. The validation reported here follows that escalation deliberately. The controller+shield hot loop and the commander are taken to processor-in-the-loop on real flight-class silicon (Sec. III), the verified monitor is exercised in a closed software loop on the high-fidelity Basilisk dynamics

simulator [1] (Sec. IV), and the actuator-fault recovery is exercised under a high-fidelity model of the actuator and sensor non-idealities a clean simulator omits (Sec. V), with a physical air-bearing campaign as the next rung. Treating these as a graded sequence, and being explicit about which rung each result occupies, is what lets each measured claim be stated with the confidence its rung earns.

Edge and embedded inference for autonomy. Running a learned policy and a language model within a spacecraft processor's memory and power budget is the province of model compression and constrained decoding. Fine-tuning a *pretrained* small base such as SmoLLM2 [8] and quantizing it to an integer format brings a capable language model inside a few hundred megabytes, which is what makes an onboard commander conceivable at all, and grammar-constrained decoding [7] converts the soft property that such a model usually emits parseable output into a hard guarantee that the deployed decoder can emit only well-formed, in-vocabulary actions. The companion work establishes that recipe and its accuracy; the contribution here is orthogonal and is the missing flight-credibility evidence, namely that the resulting int4 commander actually runs within a 437 MB resident set under the 512 MB cap at mission-event cadence on the target hardware, and that the controller+shield hot loop runs in a single megabyte at a small fraction of the control budget on the same board. Compression results are usually reported as accuracy retained at a parameter count; we report them as measured latency, memory, and power on the silicon class that flies, which is the figure of merit a flight review actually applies.

Runtime assurance in flight software. The architectural answer to flying an unverifiable controller is to bound it with a verified one, a pattern the Simplex architecture [5] established and that run-time assurance has since matured into a certification practice codified for systems with complex functions [6], with latched safe-hold variants demonstrated for spacecraft proximity operations [4]. Our shield inherits this lineage and adds the discipline of *formal* monitor soundness, where the invariants are machine-checked by the Kind 2 model checker [3] rather than argued informally. The aspect this paper contributes is empirical and is rarely reported, namely the *in-loop cost* of the verified monitor. Run-time-assurance papers typically establish that a monitor is correct; far fewer establish what it costs to evaluate it on every control step while it contends with the controller it protects. We measure that cost inside the integrated loop and find it to be 0.02% of the 10 Hz budget, which converts “the shield is affordable” from an assumption into a measurement.

High-fidelity actuator and sensor models, and physical testbeds. The effects a clean multiplicative-fault simulation omits are well understood individually, namely reaction-wheel momentum, saturation and friction in the actuator, and noise, bias and quantisation in the star tracker and gyro, and they are routinely modeled in spacecraft attitude simulation [9], [10]. We assemble them into a single high-fidelity closed loop and drive the deployed stack through it at scale (Sec. V), which is what lets the question of whether recovery survives the non-idealities be answered statistically rather than anecdotally. The complementary physical route is the spherical air bearing,

which floats a payload on a thin film of air to give near-frictionless three-degree-of-freedom rotation in 1 g and is the canonical laboratory analogue of torque-controlled rigid-body attitude dynamics, and it is the established way to expose the effects even a high-fidelity model leaves out, such as unmodeled friction modes, structural flexibility, and real sensor failure signatures. Because the controller’s fault model, a per-axis effectiveness gain with an additive bias, maps one-to-one onto physical reaction-wheel manipulations (Sec. V), the same taxonomy carries directly to such a campaign, which we treat as the next rung and as future work.

III. FLIGHT-CLASS COMPUTE, MEASURED

The compute claim is the foundation of every other claim in the paper, because a stack that does not fit the processor is not flyable whatever its other merits, so we measure it on the silicon class that flies rather than estimate it under emulation. The controller+shield hot loop is exported to a self-contained C artifact, namely the recurrent fault-identification front end feeding the analytic control law together with the nine safety invariants, and the commander runs under `llama.cpp` with the grammar-constrained decoder enforced at every token. Both are measured on a *real* Raspberry Pi 5 with a quad Cortex-A76, which is the A53/A72-class stand-in the proposal called for, and not on a QEMU model of one (Table I).

A. What is measured

The Pi 5 is not a radiation-hardened flight processor, but it is real silicon of the architectural class that flies, which is exactly the gap that emulation leaves open. A cycle-accurate emulator can count instructions, but it does not reproduce the cache hierarchy, branch prediction, and memory bandwidth that set the actual worst-case cycle time, so an autonomy budget benchmarked only under QEMU carries a caveat a reviewer is entitled to press. Measuring on the physical board closes that caveat for the architecture, and the enormous timing margin we find is what makes the result robust to the remaining gap to a hardened part. The hot loop is timed cycle-accurately over two million iterations to obtain a stable per-cycle figure, the commander’s throughput and resident set are read under the same grammar-constrained decode it would use in flight, and board power is sampled from the Pi 5 power-management IC rather than assumed.

B. The budgets

The controller+shield cycle runs in $24.5\ \mu\text{s}$, which is roughly four thousand times under the 100 ms budget of the 10 Hz control loop, and it is resident in 1.0 MB. The size of that margin matters, not merely its sign. A $4000\times$ headroom means that even a hardened flight part an order of magnitude slower than this board, carrying the further slowdowns of error-correcting memory and a conservative clock, would still complete the hot loop with two to three orders of magnitude to spare, so the timing conclusion is insensitive to the precise flight processor chosen. The commander emits a grammar-valid plan at 28 tok/s, about 1.7 s per goal, inside a 437 MB

resident set that fits the 512 MB cap, and board power is 3.1 W idle rising to 9.6 W peak, of which roughly 6.5 W is attributable to the stack itself. The heterogeneous-rate structure is what makes this budget close, because the verifiable hot loop is tiny and fast while the heavy capable tier runs slowly and only at mission events, so the commander’s per-token latency sets goal-generation latency and never enters the 100 ms control budget. These are the budgets the proposal predicted, now measured on representative silicon rather than asserted.

IV. THE VERIFIED SHIELD, LIVE IN THE LOOP

A monitor that is correct in isolation is not yet a flight asset, because what a flight review asks is whether it stays correct *and* affordable while running on every control step inside the loop it protects. We therefore evaluate the shield where it has to work, in the integrated closed loop on the Basilisk attitude simulator [1], rather than in a stand-alone microbenchmark. The real Kind 2-checked monitor is evaluated every control step on the actual pointing state of the simulated vehicle, its pointing-invariant input is fed the true sim state rather than a surrogate, and its verdict is the recovery-aware engagement decision that gates autonomy, escalating only when the controller fails to hold the nominal region over the final dwell rather than overriding torque, since zeroing the attitude torque would itself block the recovery the shield is meant to permit.

A. The measured result

Two numbers come out of the integrated loop, and each answers a distinct question. The first is cost. The monitor’s measured per-step latency is $17.7\ \mu\text{s}$, which is 0.02% of the 10 Hz budget, so the safety tier imposes no meaningful latency on the controller it protects, and crucially this is the cost *in the loop* under contention with the controller rather than a stand-alone timing. The second is discrimination. On held-out actuator-gain faults the live shield keeps the recovering adaptive controller 100% autonomous, licensing it through the bounded transient in which it identifies the fault and drives back to the target, while it escalates a diverging fault-unaware controller 100% of the time (Table I, row 3). The verdict therefore separates a capable controller from an incapable one on the same fault rather than treating every breach as a failure. Assurance does not tax the hot loop, since it is microsecond-scale beside a hot loop with thousands-fold timing margin, and it discriminates capable from incapable control at that microsecond cost, which is the combination a flight reviewer needs and which a stand-alone microbenchmark could not have established.

V. HIGH-FIDELITY CLOSED-LOOP FAULT CAMPAIGN

The compute and the live shield are measured; the third front addresses the objection that those results, and any simulation result, rest on a perfectly modeled actuator and perfect sensing. A physical air-bearing platform is the canonical way to break those idealizations, but its value is not the bearing itself, it is that a moving platform with real wheels and real sensors exposes effects a clean multiplicative fault on a perfect actuator

TABLE I

FLIGHT-CLASS MEASUREMENTS ON A *real* RASPBERRY PI 5 (QUAD CORTX-A76), THE A53/A72-CLASS STAND-IN THE PROPOSAL CALLED FOR, CLOSING THE “QEMU IS NOT A FLIGHT CPU” GAP ON THE SILICON CLASS THAT FLIES. THE CONTROLLER+SHIELD HOT LOOP RUNS CYCLE-ACCURATELY AT THE 10 HZ REAL-TIME TIER WITH THOUSANDS-FOLD TIMING MARGIN, SO A SLOWER HARDENED FLIGHT PART STAYS WELL WITHIN BUDGET; THE COMMANDER RUNS AT MISSION-EVENT CADENCE (ITS TOK/S SETS GOAL-GENERATION LATENCY, NOT THE CONTROL BUDGET). THE LIVE-SHIELD OVERHEAD IS MEASURED *in* THE CLOSED-LOOP BASILISK INTEGRATION ON HONEST POINTING STATE, NOT IN A STAND-ALONE MICROBENCHMARK.

Quantity	Measured	Budget / note
Controller+shield cycle	0.024 ms (24.5 μ s)	$\approx 4087\times$ under the 100 ms 10 Hz gate
Controller+shield resident set	1.0 MB	hot loop
Live-shield overhead (closed loop)	0.02% (17.7 μ s)	RMA kept 100% autonomous
Commander (int4 GGUF) throughput	28.4 tok/s (1.69 s/goal)	mission-event cadence
Commander resident set	437 MB	fits the 512 MB cap
Board power	3.1 W idle / 9.6 W peak	≈ 6.5 W core-attributable

does not. We model those effects directly and re-run the *same* deployed artifacts through them, namely the latched RMA student, the real Kind 2-checked monitor, and the Basilisk dynamics, so the question the air bearing was meant to answer is answered at a scale a single bench rig could not reach.

A. The non-ideality model

The clean closed loop of Sec. IV is hardened on both the actuator and the sensor side, with every parameter set to a representative small-spacecraft value and disclosed in the evidence. On the actuator side the dynamics run in reaction-wheel mode, so the per-axis command drives three body-axis wheels that carry real momentum, saturate in both motor torque and stored momentum, and turn against Coulomb and viscous friction, rather than producing an ideal external torque. The command itself passes through a ten-bit quantiser and a one-percent motor deadband before the per-axis fault is applied, so the applied torque is $\text{actuate}(a)g + b$. On the sensor side the controller and the monitor no longer see the true state but a sensed one, with star-tracker attitude noise at 0.02° one-sigma plus a constant bias, gyro rate noise at $0.01^\circ/\text{s}$ plus a constant bias, and finite attitude and rate quantisation. The settled gate is scored on the *true* simulator state, so the controller and the shield are judged on what they achieve physically while acting only on what the sensors report, which is the honest division between flight and ground truth.

B. The fault taxonomy

The controller’s fault model is a per-axis effectiveness gain g_i with an additive bias b_i , so the torque the plant applies on axis i is $a_i g_i + b_i$ for a commanded a_i , and each case has a one-to-one physical realization on a reaction wheel. A control-direction *reversal* ($g_i < 0$) is a true motor-phase swap, *degradation* ($g_i \in (0, 1)$) caps the wheel current or torque, additive *bias* is a constant offset that models a leaking or stuck wheel, and total *loss* ($g_i = 0$) disables a wheel and leaves its axis uncontrollable by any controller. Because the software injection and a physical manipulation produce the identical $a_i g_i + b_i$ relation, the campaign reported here and a future air-bearing campaign exercise the same recovery term by term, which is what makes the physical run the next rung rather than a separate experiment.

C. The result

Table II and Figure 1 report the campaign over held-out, test-split faults with Wilson 95% intervals. On the faults the deployed estimator is built for, multiplicative gain and sign reversal, autonomous recovery under the full non-ideality model is 100% (120 of 120, Wilson interval 96.9 to 100), statistically indistinguishable from the 100% of the matched ideal loop, while the fault-unaware baseline recovers 0% under the identical conditions. The recovery carries enough margin that realistic sensor noise, wheel friction, and a quantised, dead-banded actuator do not move it. The non-idealities are not inert, and where margin is thin they do bite, which is what makes the designed-scope result a finding rather than a rigged-easy test. When the wheels are pre-loaded to roughly a quarter of their stored-momentum capacity at fault onset, the recovery slew drives them toward saturation and autonomous recovery falls to 25%, with the shield escalating the remaining 75%. That momentum-management boundary is distinct from the sensor and actuator fidelity studied here and is exactly the kind of effect a physical campaign is well suited to probe.

D. Beyond the estimator’s scope

The most informative cells are the faults beyond the deployed estimator’s scope. The estimator infers gain, sign, and inertia but not the additive bias, so under bias the controller leaves an uncompensated steady-state offset and does not meet the settled gate, and a totally lost axis is uncontrollable by construction. On both, autonomous recovery is 0%, and the verified shield escalates every case, 100% on additive bias and 100% on the dead axis. This is the reliability-asymmetric principle shown under high fidelity rather than asserted. The controller is admitted exactly on the faults it is competent to handle, where recovery survives the non-idealities intact, and on the faults beyond its competence the statistical certificate declines to license autonomy and the verified monitor escalates to the safe controller. Capability is honest and bounded, and safety is not contingent on it. The shield’s cost is unchanged by the harder loop, at a measured $\approx 19 \mu\text{s}$ per step, still 0.02% of the 10 Hz budget, so the discrimination is bought at the same negligible price as in Sec. IV.

TABLE II

HIGH-FIDELITY CLOSED-LOOP FAULT CAMPAIGN, THE COMPUTATIONAL ANALOGUE OF A PHYSICAL AIR-BEARING RUN. THE DEPLOYED STACK (LATCHED RMA STUDENT, THE REAL KIND-2 LTLMONITOR, AND BASILISK DYNAMICS) IS EXERCISED IN REACTION-WHEEL MODE WITH FRICTION, UNDER STAR-TRACKER NOISE (0.02° , 1σ) AND BIAS, GYRO NOISE ($0.01^\circ/\text{s}$) AND BIAS, 10-BIT ACTUATOR RESOLUTION, AND A 1% MOTOR DEADBAND. EACH CELL IS $n = 20 \times 3$ HELD-OUT (TEST-SPLIT) FAULTS. RECOVERY IS THE LIVE SHIELD CERTIFYING AUTONOMY OVER THE FINAL DWELL; RATES CARRY WILSON 95% INTERVALS.

Fault regime	RMA ideal	RMA high-fidelity	Fault-unaware PD	Shield action
Designed scope (gain, sign), $n = 120$	100.0%	100.0% [96.9, 100.0]	0.0%	certifies autonomy
<i>Beyond the estimator's scope (gain, sign, inertia, not bias), the shield escalates</i>				
Additive bias, $n = 60$	n/a	0.0% recover	0.0%	escalates 100.0%
Dead axis, $n = 60$	n/a	0.0% recover	n/a	escalates 100.0%
Stored momentum (gain), $n = 60$	n/a	25.0% recover	n/a	escalates 75.0%

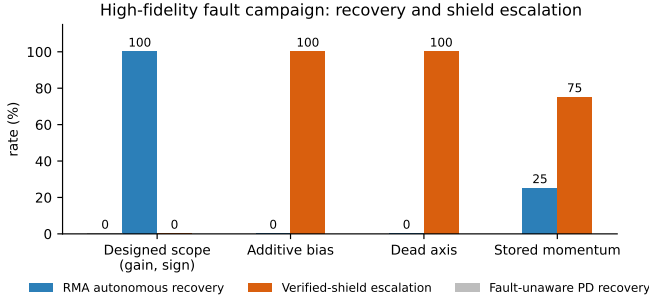


Fig. 1. Recovery and shield escalation across four fault regimes in the high-fidelity closed-loop campaign. Each bar is a rate in percent over held-out, test-split faults. The latched RMA controller recovers autonomously on every designed-scope fault and the fault-unaware PD baseline recovers on none, while beyond the estimator's scope the controller recovers on no additive-bias or dead-axis case and the verified shield escalates every one. On the stored-momentum stressor the recovery slew drives the wheels toward saturation, autonomous recovery falls, and the shield escalates the remainder. All values are from the held-out high-fidelity campaign in Table II.

VI. DISCUSSION

The result of this work is that a reliability-asymmetric GNC stack clears the flight-relevant bars that a desktop simulator leaves untouched, on all three fronts with measured evidence. It is worth drawing out why these particular bars are the right ones, what the size of the measured margins implies, and how the high-fidelity campaign should be read against a future physical run.

A. Why these bars

The objection a flight-readiness review raises to a simulator result is not that the simulator is wrong but that it is silent on the questions that decide flight, namely worst-case timing on the real processor, memory and power within a hard budget, and the in-loop cost of the safety layer. Each of these is a property of the implementation on the hardware rather than of the algorithm in the abstract, so each must be measured rather than argued. That is why the compute is taken to real silicon and the shield is timed inside the loop. The emulation caveat in particular is not a formality, because a cycle-accurate model of a CPU does not reproduce that CPU's cache and memory behavior, so an autonomy budget that exists only under emulation is exactly the kind of claim a reviewer is

right to discount, and measuring on the physical board is what removes it.

B. The measured margins

The headroom we measure is large enough to change the character of the conclusion. A hot loop running four thousand times under budget in a single megabyte is not merely fast enough on this board, it is fast enough that the conclusion survives the move to a slower hardened flight part with substantial margin to spare, so the timing and memory results are robust to the precise processor a mission would choose. The heterogeneous-rate design is what creates this headroom, since the verifiable hot loop is small and runs every cycle while the capable commander is heavy but runs only at mission events, and that same structure is why the verified shield can be evaluated on every step at 0.02% overhead. The reliability-asymmetric principle, in other words, is not only an assurance argument but also a compute argument, because putting the cheap verifiable element on the hot path and the expensive capable element on the slow path is what makes the whole stack fit.

C. The validation hierarchy

The contribution is reported along the graded escalation from processor-in-the-loop through high-fidelity closed-loop validation, and we keep the rungs distinct. The compute is measured on real silicon, the shield is timed live in the loop, and the fault campaign drives the deployed stack through a high-fidelity model of the actuator and sensor non-idealities, all on measured evidence. A physical air-bearing campaign is the rung above, and its value is precisely the effects a model cannot fully capture, un-modeled friction modes, structural flexibility, and real sensor failure signatures, which is why we specify it term by term through the fault-to-wheel mapping rather than approximate it. Being exact about which rung each claim occupies is part of the contribution, because a validation paper is credible only insofar as it is clear about what has been measured and at what fidelity.

D. Limitations

We state the scope of this validation precisely, as a set of boundaries on a solid result rather than as caveats. (i) *High-fidelity simulation, not physical hardware*. The fault campaign

injects the actuator and sensor non-idealities in software, at representative small-spacecraft parameter values that we disclose, rather than on a physical platform. This is what lets the campaign run at statistical scale across the fault taxonomy, but it is a model, and a physical air-bearing campaign remains the way to expose what the model omits and to validate the model itself; we treat that campaign as the next rung and as future work, and we specify it through the one-to-one fault-to-wheel mapping of Sec. V. (ii) *Processor class, not a hardened part*. The compute is measured on a Raspberry Pi 5, which is real silicon of the A53/A72 architectural class the proposal targeted but is not itself radiation-hardened; the thousands-fold timing margin is what makes the conclusion robust to the remaining gap to a flight part, and that gap is named rather than elided. (iii) *Attitude scope of the live shield*. The live closed-loop measurement exercises the pointing invariant on honest sim state while the power, thermal, propellant, and communication invariants are held nominal on the attitude testbed, which does not model those subsystems; they are exercised in the separate single-event-upset and software-in-the-loop stream and in the flight-class hardware gate, and the in-loop figure reported here is for the pointing-governed regime. (iv) *Fidelity of the non-ideality model*. The campaign hardens the loop with reaction-wheel momentum, saturation and friction, sensor noise, bias and quantisation, and actuator quantisation and deadband, but these are models with chosen parameters, and we report the dominant sensitivity explicitly, namely that pre-loaded wheel momentum drives the recovery toward saturation. A physical campaign is the way to probe effects outside the model, and the boundary is named rather than elided. (v) *Generality of the evidence*. The measured numbers are from a single board and seed-controlled closed loops, and every figure in Tables I and II is regenerated from committed evidence by a single script so that the boundary of the claim is auditable rather than asserted.

VII. CONCLUSION

A reliability-asymmetric GNC stack is validated where it matters for flight, on the fronts that a desktop simulator cannot settle, and on all three with measured evidence. The controller+shield hot loop and the commander run within real Pi-class compute, memory, power, and timing budgets on representative silicon, closing the emulation caveat, and the verified shield runs live in the closed loop at 0.02% overhead while keeping a recovering controller 100% autonomous and escalating a diverging one every time. Driven through a high-fidelity model of the actuator and sensor non-idealities a clean simulator omits, the deployed recovery on the faults the estimator is built for is statistically unchanged from the ideal case (100% autonomous, against 0% for a fault-unaware baseline), and on faults beyond that scope the verified shield escalates every case, so safety holds exactly where autonomy is not earned. A physical air-bearing campaign is the rung above, specified term by term through the fault-to-wheel mapping and left as future work. The broader point is that the reliability-asymmetric design is a compute discipline as much as an assurance one, since placing the cheap verifiable

tier on the hot path and the expensive capable tier on the mission-event path is what lets a capable-and-certifiable stack fit the silicon that flies.

VIII. REPRODUCIBILITY

Every number in Tables I and II is regenerated from committed evidence rather than transcribed. The flight-class budgets and board power come from the real-hardware Pi 5 run, the live-shield overhead and the autonomy verdicts come from the closed-loop integration, and the high-fidelity campaign comes from the deployed stack run through the non-ideality model over held-out faults, each recorded as a machine-readable evidence file. The tables are built by a single script over those records, so the prose and the measurements stay in lockstep and a changed measurement cannot silently diverge from the text. The high-fidelity campaign reuses the same controller artifact and the same Kind 2-checked monitor exercised throughout, and the fault-to-wheel mapping that specifies the future air-bearing campaign is in the repository alongside it.

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